Presentation Speaker Notes & Script

Forecasting U.S. Manufacturing Stress with a Governed Adaptive Agent

TARGET DURATION	FORMAT	FOCUS AREA
7 – 10 Minutes	**2 Presenters**	**Architecture & Governance**

PART 1: PRESENTER 1 — PROBLEM, SOLUTION & ARCHITECTURE

Slide 1 — Title Page ~25s

> *"Good morning/afternoon. We're Team BMO Two, and today we're presenting **Forecasting U.S. Manufacturing Stress with a Governed Adaptive Agent**."*
>
> *"Up front, our honest framing: this was a learning exercise to design, evaluate, and govern an agentic forecaster—not to ship an immediate production model. Please keep that lens on everything we show today."*

Slide 3 — Business Problem & Opportunity ~45s

> *"Why does this matter? Credit teams want earlier warnings when an industry heads into stress. But manufacturing stress is rare—occurring in only **20 of 320 months** between 2000 and 2026, clustered around 2008–09, 2020, and brief 2021–22 episodes."*
>
> *"Our target definition: a month is stressed when U.S. manufacturing production—the FRED **IPMAN** series—falls **2% or more over the previous 3 months**, forecasted **3 months ahead**. Having few events combined with hard-to-audit analyst judgment is precisely where a governed agent can help."*

Slides 4 & 6 — Our Solution & Architecture ~45s

> *"Our solution is a workbench pairing a transparent statistical model with an LLM analyst. It has three core components:"*
>
> *1. A **leakage-safe numerical anchor** using logistic regression refit at each date with historical data only.*
> *2. A **bounded LLM analyst** that suggests a probability and rationale, but whose adjustment to the anchor is **capped at ±0.03**.*
> *3. A **governed adaptive memory** where hypotheses require 3 distinct confirmations before altering strategy, with every change hash-linked."*
>
> *"Crucially, the agent can adjust probabilities and strategy text, but **cannot** alter underlying data, features, or anchor code. No retraining happens through the agent."*

Slides 8, 10 & 11 — Technical Approach & Backtest Results ~60s

> *"To test this, we used **15 monthly public inputs** across **84 monthly backtest origins and 6 stress events**."*
>
> *"The headline finding across models? **No model beat the historical-frequency baseline (Brier score 0.0667)**. Our bounded adaptive agent came in close at **0.0692**, while the unbounded LLM analyst performed poorly at **0.0979**. Why? Stress events are rare, and macro signals are largely coincident rather than leading three months out. I'll now hand over to Presenter 2."*

PART 2: PRESENTER 2 — GUARDRAILS, GOVERNANCE & LEARNINGS

Slides 13 & 14 — Guardrail Behavior & LLM Comparison ~45s

> *"Thanks Presenter 1. Did our safety guardrail work? Absolutely. The adaptive agent frequently wanted to be much more aggressive—proposing an average probability of **0.094** compared to the anchor's **0.071**."*
>
> *"However, our **±0.03 cap** engaged on 7 critical origins—including mid-2020 when the agent proposed up to **0.85**—pulling the final forecast back down. The guardrail successfully **limited harm**, keeping our bounded agent close to the baseline while unbounded variants faltered."*

Slide 15 — Governed Adaptive Memory ~40s

> *"We also tested outcome-grounded learning with strict temporal control. Outcomes stay hidden until their official release date, entering a resolution ledger only when published."*
>
> *"A dedicated reviewer evaluates resolved origins and updates a versioned strategy file (`skill_state.yaml`). Every single update records timestamps, tools, and before/after hashes for complete traceability."*

Slide 17 — Key Takeaways ~45s

> *"So, what are our main takeaways from this project?"*
>
> - ***One:** Always start with the base rate—for rare events, it is a fiercely demanding benchmark.*
> - ***Two:** Anchor and bound your LLMs; left unbounded, they tend to overreact.*
> - ***Three:** Signals matter more than model complexity; better leading indicators beat a bigger model.*
> - ***Four:** Governance can—and should—be proven before raw predictive accuracy is achieved.*

Slides 18 & 23 — Next Steps & Closing ~30s

> *"Where do we take this next? We want to run a frozen prospective test with immutable pre-outcome logging, incorporate forward-looking leading indicators like new orders or surveys, and refine explanation quality."*
>
> *"The ultimate value here isn't just a forecasting hack; it's a **reusable governance pattern for regulated financial environments**—combining a statistical anchor, a bounded agent, and a complete audit trail. Thank you!"*

Q&A PREPARATION CHEAT SHEET

Q1: "If nothing beat the base rate, did the project fail?"

Answer: No. The primary deliverable was to demonstrate **safety, evaluation rigor, and governance**. For rare events, beating a historical base rate is notoriously difficult. We proved that a leakage-safe framework can bound an LLM to prevent severe overreaction while maintaining transparency.

Q2: "Why was the guardrail cap set specifically at ±0.03?"

Answer: It was chosen as a conservative parameter to keep the agent tightly tethered to the statistical anchor. Backtest results confirmed its necessity: without the cap, the unbounded agent overreacted to noise and scored significantly worse (Brier score 0.0979 vs 0.0692).

Q3: "Did the agent actually learn over time?"

Answer: While raw forecasting performance didn't surpass the baseline, the agent proved out an **auditable learning workflow**. The review mechanism successfully executed 35 mutations across 24 reviews under strict governance